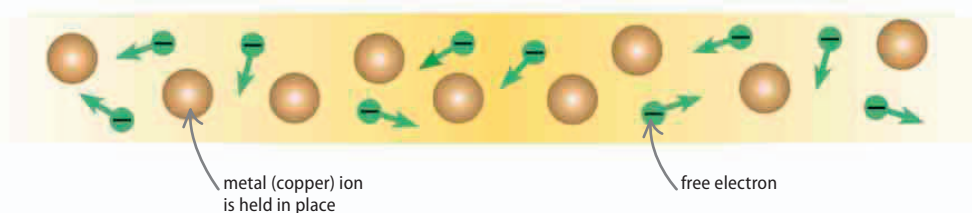


Electric currents

When electric charge flows through a material, it is called an electric current. It is caused by a drift of electrons through a material called a conductor (see below). In an electrical circuit (see page 206), a power source, such as a battery, gives the electrons energy so that the charge flows from the negative terminal (connection) of the power source around the circuit to the positive terminal. Current only flows when an electric circuit is complete, with no gaps. In a circuit, individual electrons actually travel extremely slowly (less than 1 mm (0.04 in) per second), but because they are closely situated to one another they are able to pass electrical energy around the circuit at more than 100 million m (328 million ft) per second.

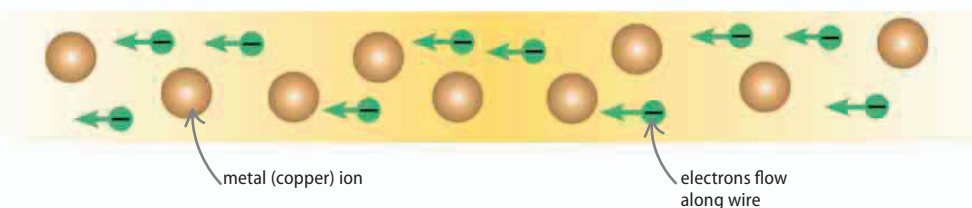
▷ Disconnected conductor

If a wire is not connected to a battery, the free electrons within it move randomly in all directions.



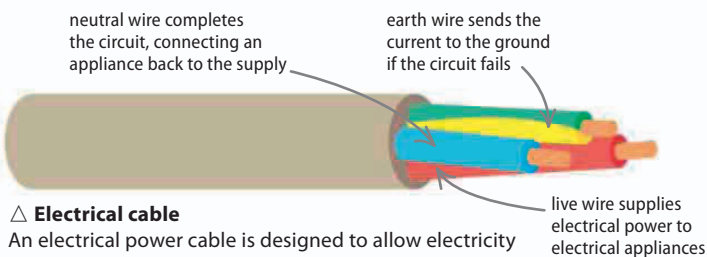
▷ Current-carrying conductor

When the conductor forms part of a powered electrical circuit, the electrons drift toward the positive pole of the power supply.



Controlling electricity

Some materials are better at carrying an electrical current than others and are called conductors. Many metals make good conductors, as their atoms easily release electrons to carry the current. Materials such as glass, rubber, and most plastics are made of atoms that do not easily release their electrons. As a result, these conduct electricity poorly, cannot carry a current, and are called insulators.



△ Electrical cable

An electrical power cable is designed to allow electricity to flow easily along its copper wires. Each wire is separated by a plastic sleeve—a good insulator. The colors of the sleeves vary between countries.

REAL WORLD

Amber

Amber is the dried resin of certain trees, and it quickly collects a static charge when it is rubbed. A piece of charged amber will attract light objects, such as feathers. The ancient Greeks were aware of these effects, and the words “electron” and “electricity” come from the Greek word for amber.

